

Federated LoRA Fine-Tuning of Language Models for Network Intrusion Detection on NF-ToN-IoT

Interim report · 16 September 2026 · 7 of 8 runs complete

Experiment

Project	fed-llm-ids
Tracking	comet.com / text-films / fed-llm-ids
Code	github.com/asfi50/Fed_GNN (branch: llm)
Adapters	huggingface.co / fed-ids-<model-name>

Environment

Hardware	Kaggle notebook · NVIDIA Tesla T4, 16 GB
Software	transformers 5.17.0 · peft 0.21.0 · torch 2.10.0+cu128
Precision	bert, t5: fp32 · llama, qwen: fp16 base with fp32 adapters
Seed	42 (partitioning, sampling and initialisation)

Models

Checkpoint	Family	Parameters	Trainable (LoRA)	Share
■ bert-base-uncased	encoder	110 M	0.60 M	0.543 %
■ t5-base	encoder-decoder	225 M	1.77 M	0.785 %
■ meta-llama/Llama-3.2-1B	decoder	1,239 M	3.43 M	0.277 %
■ Qwen/Qwen3.5-2B	decoder	2,215 M	1.50 M	0.068 %

Dataset

Source	NF-ToN-IoT · netflow records · 10 attack classes
Rows	1,379,274 (used in full, no subsampling)
Validation	137,928 stratified, shared by every run
Test	137,928 stratified, shared by every run
Training pool	1,103,418 partitioned across 5 clients

Input representation

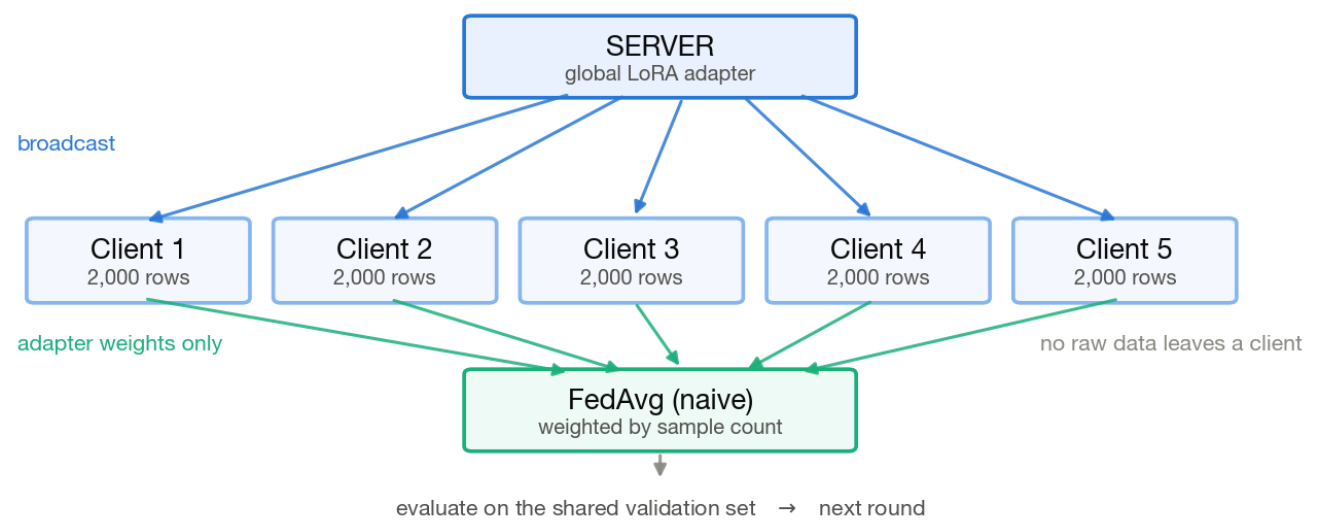
Each netflow row is serialised into a compact **key=value** string — protocol, ports, byte and packet counts, TCP flags and duration — then tokenised to at most 128 tokens.

Label (the binary attack flag) and **Attack** (the target) are dropped as outright leaks, and both IP columns with them: attacker addresses are near-unique per class in this dataset, so keeping them would let a model memorise addresses rather than learn traffic behaviour.

Scope. Naive FedAvg aggregation only. The three alternative strategies implemented in the pipeline — delta-SVD, FFA-LoRA and performance weighting — are left for a separate study.

Pipeline

One federated round, repeated 10 times.



Federated configuration

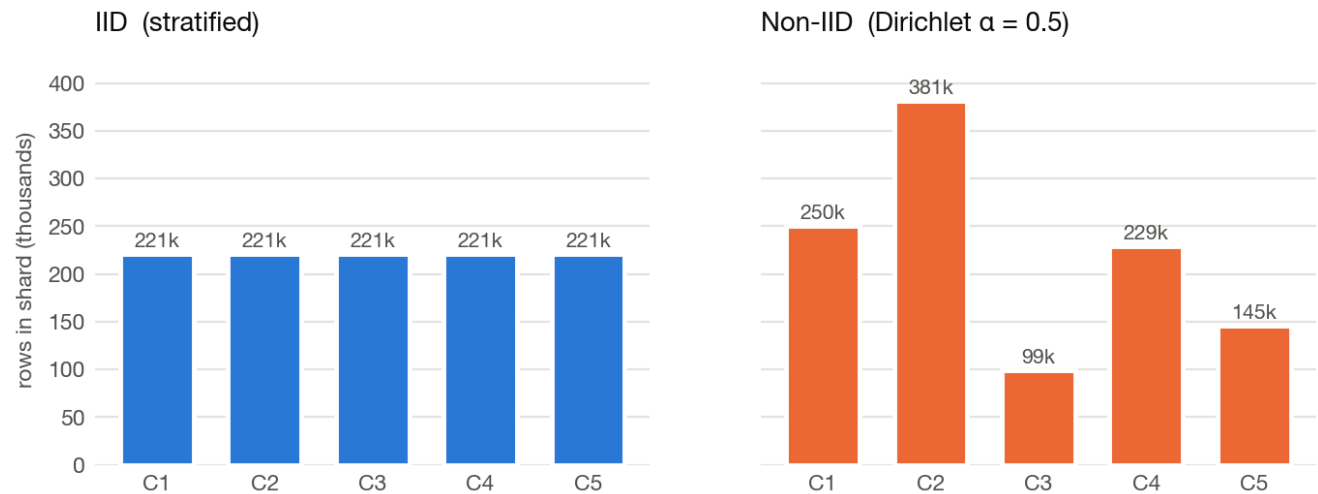
Setting	Value
Clients	5
Rounds	10
Local epochs	1 per round
Rows per client per round	2,000 — a fresh slice each round, never repeated
Rows seen per client	20,000 of its own shard
Aggregation	naive FedAvg, weighted by client sample count

Adaptation

Setting	Value
Method	LoRA · rank 16 · alpha 32 · dropout 0.05
Target modules	query, value (bert) · q, v (t5) · q, k, v, o (llama, qwen)
Optimiser	AdamW · lr 2e-4 · weight decay 0.01 · 6 % warmup
Batch size	32 (bert, t5) · 16 (llama, qwen)
Class weights	inverse frequency, clipped at 50

Client shards

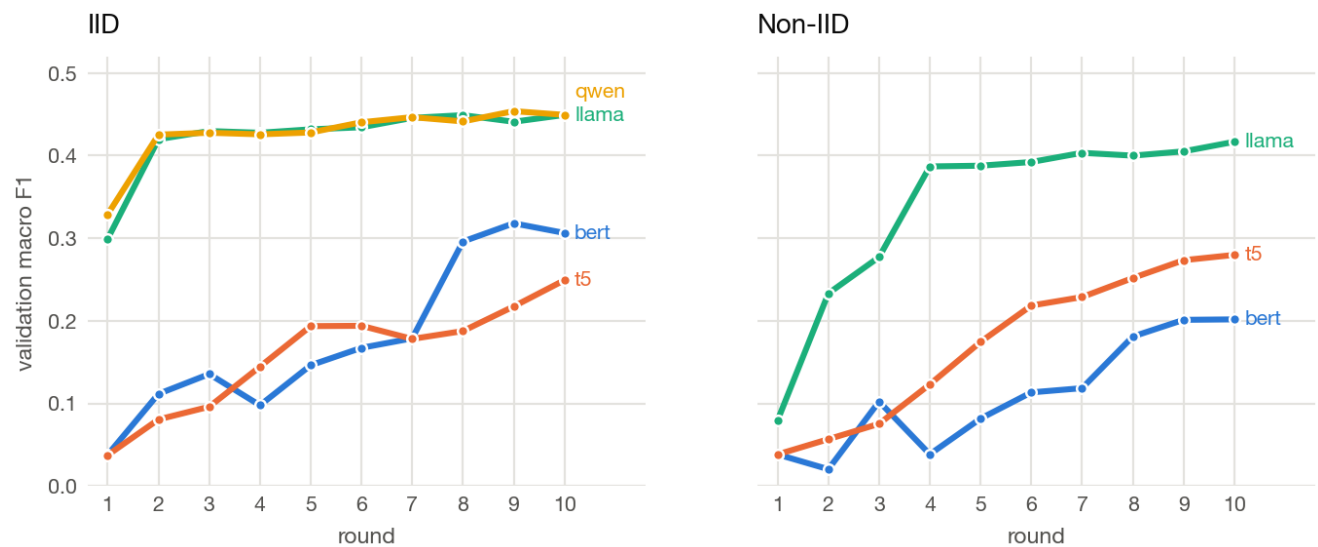
How the 1,103,418 training rows are divided between the five clients.



Each client owns a fixed private shard and walks through it in consecutive slices, one per round, so it never trains on the same row twice. The per-round sample keeps the client's own class distribution intact — rebalancing it would erase the very skew the non-IID split exists to measure. Largest-to-smallest shard ratio: $1.0\times$ IID, $3.9\times$ non-IID.

Convergence

Validation macro F1 after each aggregation round.



Every model is still climbing at round 10 — none has converged. The decoder models pull away within two rounds and hold the gap; bert and t5 are still below 0.3 when the budget runs out.

Results

Held-out test set · 30,000 stratified rows · identical for every run.

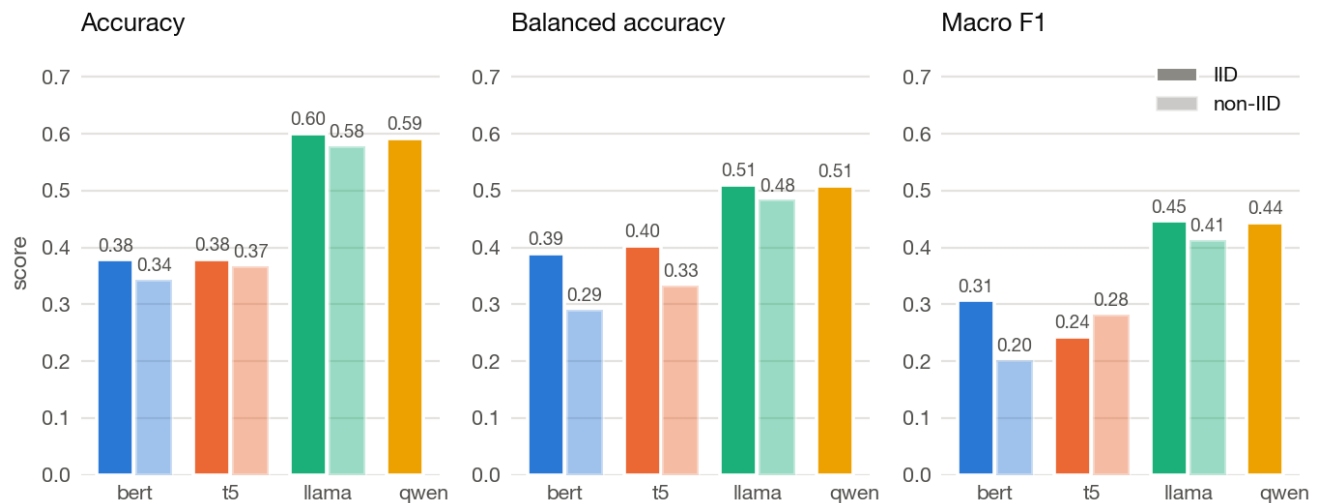
IID partition

Model	Accuracy	Balanced acc.	Macro F1	Macro F1 *	Wall clock
bert	0.378	0.389	0.307	0.383	20 min
t5	0.380	0.403	0.243	0.303	35 min
llama	0.601	0.510	0.446	0.558	32 min
qwen	0.593	0.508	0.444	0.555	88 min

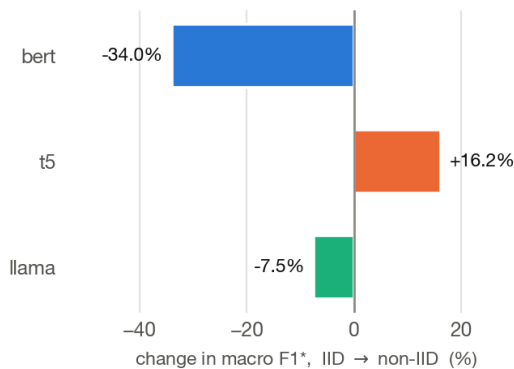
Non-IID partition (Dirichlet $\alpha = 0.5$)

Model	Accuracy	Balanced acc.	Macro F1	Macro F1 *	Wall clock
bert	0.344	0.291	0.202	0.253	20 min
t5	0.368	0.333	0.282	0.353	35 min
llama	0.579	0.484	0.413	0.516	31 min
qwen	—	—	—	—	not finished

* Macro F1 recomputed over the 8 classes excluding mitm and ransomware — see page 4.



Cost of heterogeneity

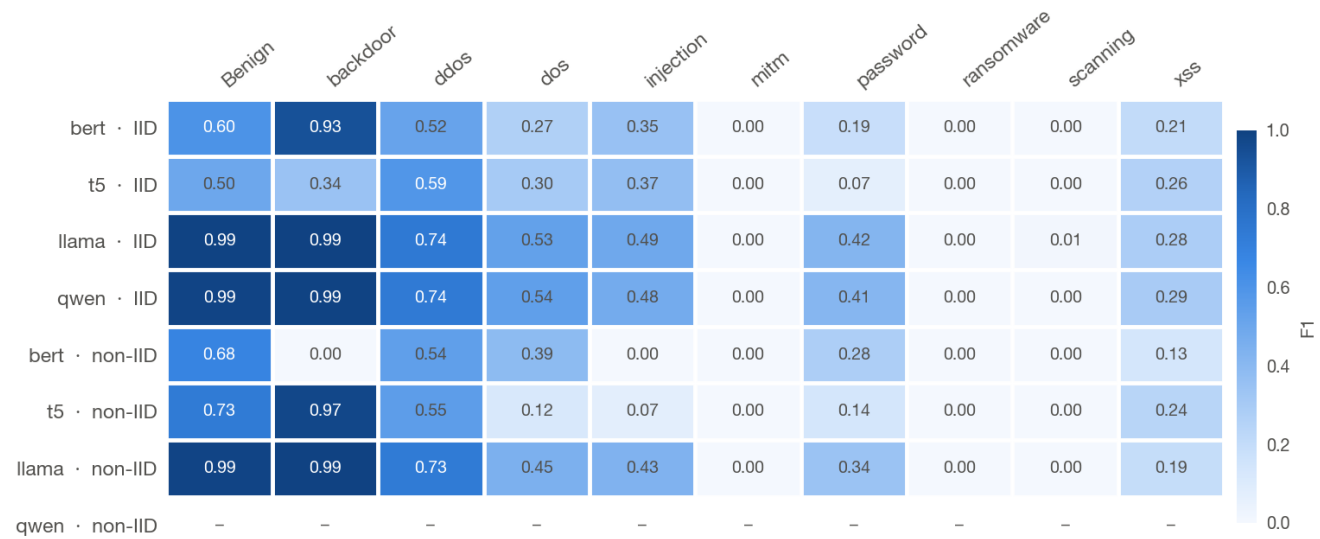


The two decoder models sit far ahead of the encoder and encoder-decoder baselines. **llama at 1.2 B matches qwen at 2 B** while taking a third of the wall clock, so the returns flatten somewhere below 2 B.

Larger models also absorb heterogeneity better: **bert gives up a third** of its macro F1 on the non-IID split, **llama under a tenth**. t5 moves the other way, but its scores are weak and unstable throughout, so the sign is not yet meaningful.

Per-class performance

F1 on the held-out test set, all models and both partitions.



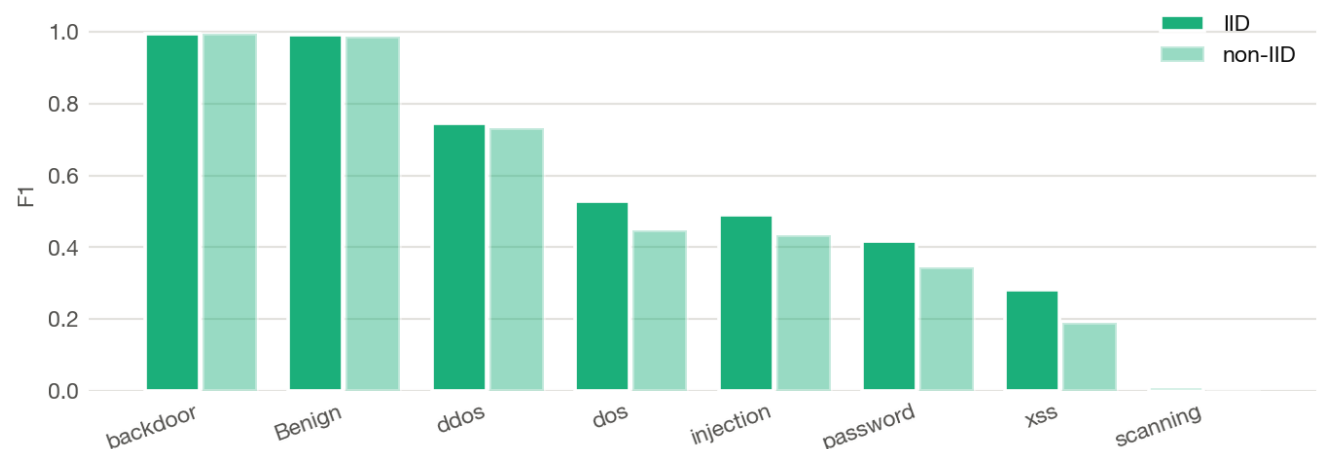
Three classes never learn

Class	Rows in dataset	Share	Rows seen / client / round	F1 (llama)
ransomware	142	0.010 %	0.2	0.00
mitm	1,295	0.094 %	1.9	0.00
scanning	21,467	1.556 %	31.1	0.01
backdoor	17,247	1.250 %	25.0	0.99
ddos	326,345	23.661 %	473.2	0.74

A client draws 2,000 rows from its shard each round. **ransomware** is 0.01 % of the data, so a client meets roughly one such row every ten rounds — far too few to learn from. **mitm** is barely better, and **scanning** is almost never predicted despite having enough examples.

Macro F1 averages over all ten classes, so these three zeros pull llama from **0.558 down to 0.446**. The GNN branch of this project already excludes mitm and ransomware; matching that choice here would make the two branches directly comparable.

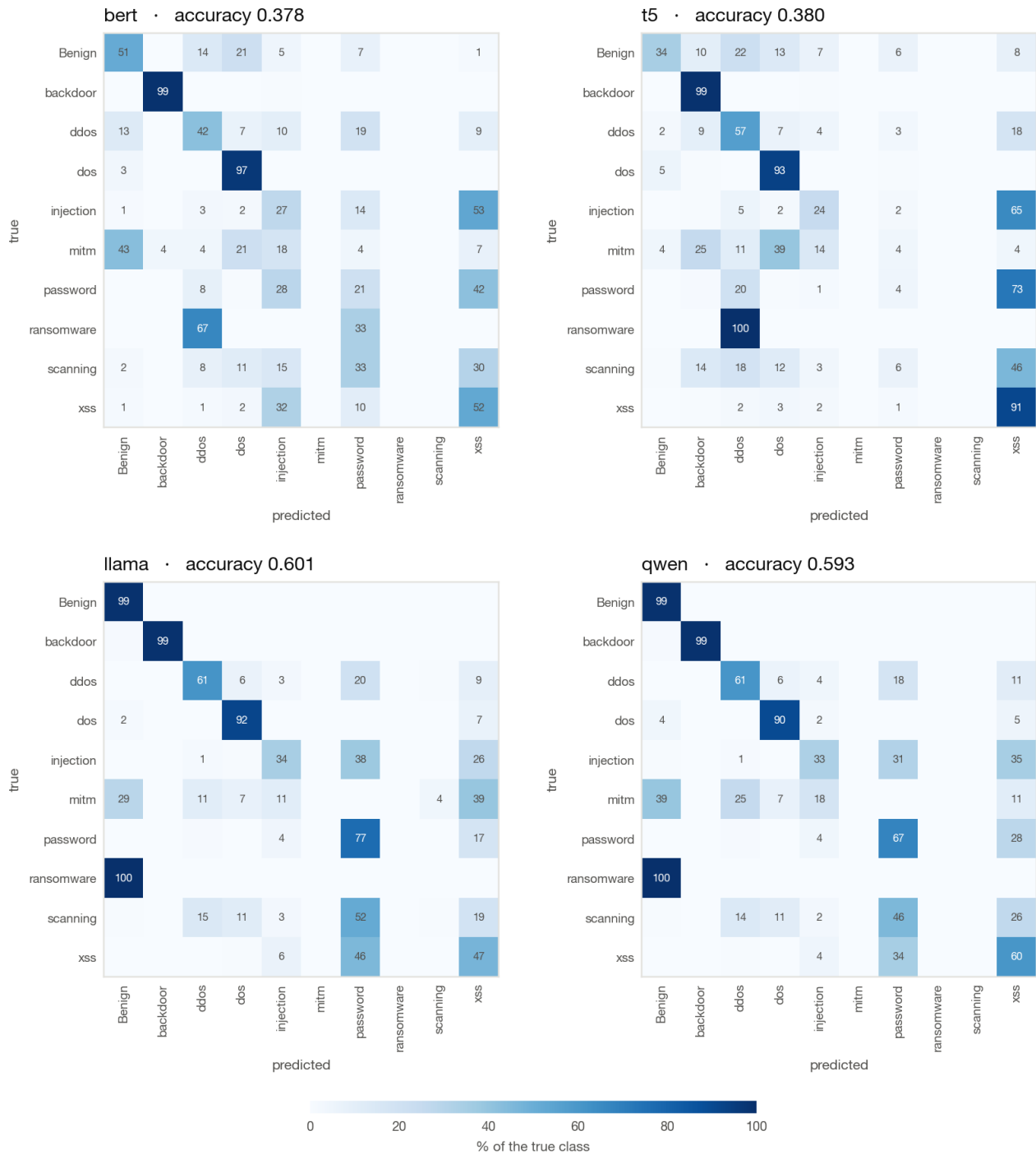
Where the strongest model still struggles



llama-3.2-1B. Benign and backdoor are essentially solved; everything below ddos degrades, and the non-IID partition costs the most on the classes that were already weakest.

Confusion matrices — IID partition

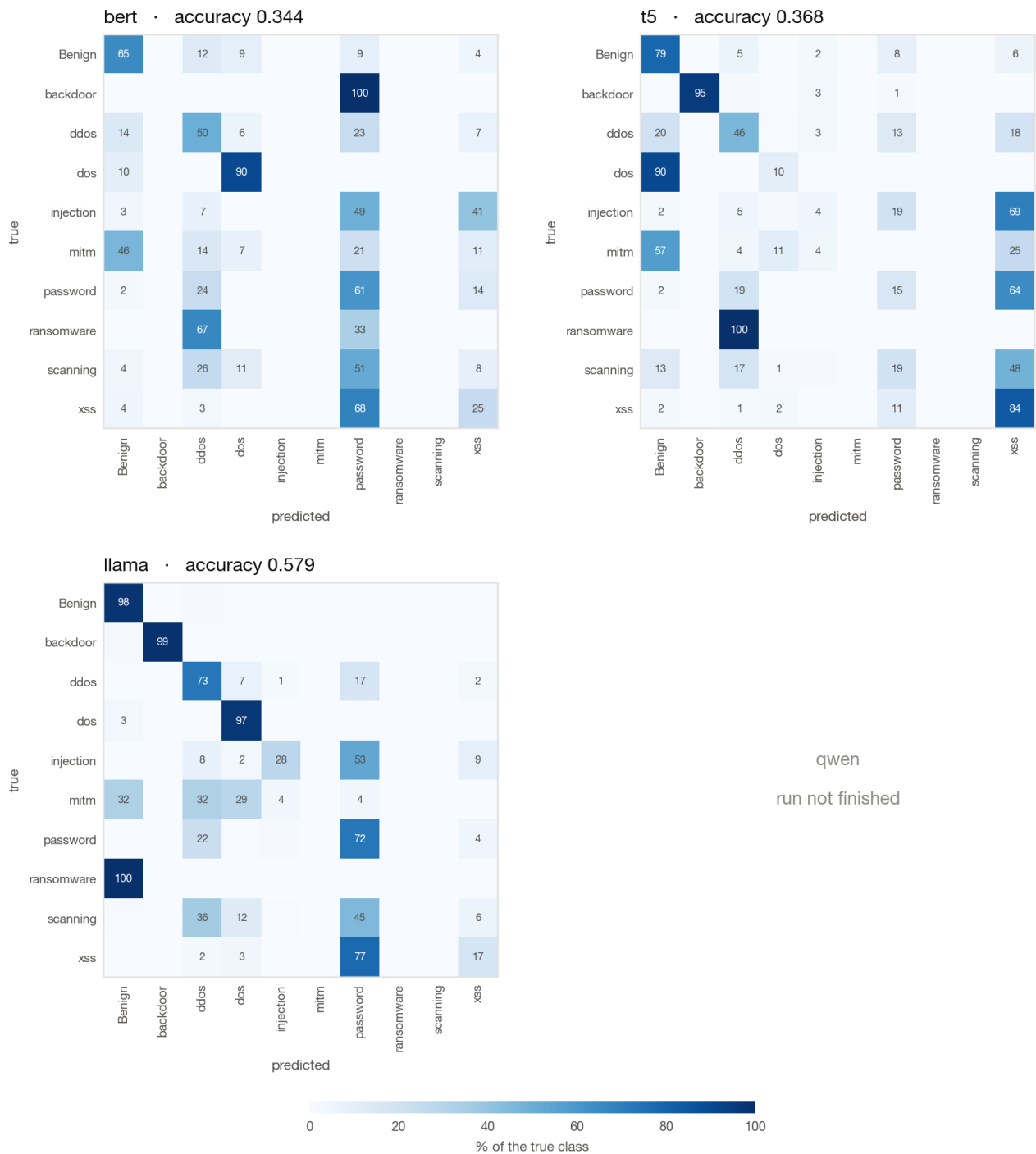
Rows are true classes, columns predicted. Each cell is the percentage of that true class sent to that column, so every row sums to 100.



A dark diagonal means predictions land on the correct class. The decoder models hold a clear diagonal for Benign, backdoor and ddos, while the lower rows stay pale because those classes are rarely predicted at all.

Confusion matrices — Non-IID partition

Rows are true classes, columns predicted. Each cell is the percentage of that true class sent to that column, so every row sums to 100.



A dark diagonal means predictions land on the correct class. The decoder models hold a clear diagonal for Benign, backdoor and ddos, while the lower rows stay pale because those classes are rarely predicted at all.

Status and next steps

Run	Status	Training	Wall clock	Communication
■ bert · IID	complete	10 rounds	20 min	114 MB
■ bert · non-IID	complete	10 rounds	20 min	114 MB
■ t5 · IID	complete	10 rounds	35 min	338 MB
■ t5 · non-IID	complete	10 rounds	35 min	338 MB
■ llama · IID	complete	10 rounds	32 min	654 MB
■ llama · non-IID	complete	10 rounds	31 min	654 MB
■ qwen · IID	complete	10 rounds	88 min	285 MB
■ qwen · non-IID	not finished	—	—	—

Outstanding

1. **qwen on the non-IID partition** — the only missing cell, roughly 90 minutes.
2. **Decide on mitm and ransomware.** Both sit below 0.1 % of the data and neither is ever predicted by any model; the GNN branch already excludes them.
3. **Centralised baselines**, to separate the cost of federation from the cost of the task itself.
4. **Aggregation study** — delta-SVD, FFA-LoRA and performance weighting, on the non-IID partition where the choice should matter most.

Open question

Absolute accuracy tops out near 0.60. Classical models on this dataset typically reach much higher, so the comparison against the GNN branch will decide how this is framed: either language models are competitive here, or they are not yet — but are markedly more robust to client heterogeneity, which stands as a finding in its own right.

Reproducibility

Every run is seeded at 42 and logs its full dataset provenance to Comet: source file, row counts, Dirichlet alpha and per-client shard sizes. Rebuilding the shards from the same seed reproduces the partition byte for byte, so nothing needs to be carried between sessions.